And thus, this brings us to the last step we need to implement: finding out if a graph is bipartite, and retrieve the two partitions.

Listing 15.6 Checking if a connected graph is bipartite

```
function
isBipartite(graph)
      #1
  queue ← new
Queue()
#2

queue.insert(chooseRandomVertex(graph))
              #3
  colors ← new HashTable()
                                                    #4
  colors[queue.peak()] ←
red                                                  #5
  while not queue.empty()
do                                                   #6
    v ←
queue.dequeue()
       #7
    for e in graph.adjacencyList[v] do
                  #8
      u ← e.dest
      if colors[u] == colors[v] then
                          #9
        return (false, null, null)
      if u not in colors
then                                                  #10
        colors[u] = (blue if colors[v] == red else red)
        queue.enqueue(u)
   return (true, {v | colors[v] == red}, {v | colors[v] ==
blue})             #11
```